One Rule to Bring Them All: Investigating Transport Connectivity in Public Transport Route Generation for Equitable Access

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Abstract

Designing a city-wide public transport network poses a dual challenge: achieving computational efficiency while ensuring spatial equity for different population groups. We investigate whether AI-based optimization — hybrid neuroevolutionary methods combining graph neural networks with evolutionary algorithms — can scale Transit Network Design Problem (TNDP) solutions from synthetic tests to real urban networks while preserving social fairness. Our contribution is to introduce a transport connectivity-aware accessibility metric that bases optimization on principles of equitable accessibility rather than traditional trade-offs between passenger and operator costs. The results show a noticeable improvement in network resilience by improving algebraic connectivity on synthetic datasets, and highlight the ambiguity of applying network generation to real data.

Introduction

To build more equitable and sustainable cities, we need to recognize the new demands posed by new urban science. Both new methods of studying and modeling the urban environment and the emergence of new forms of cities open up new perspectives for studying existing cities and assessing the consequences of new urban phenomena (Ye et al. 2025). Artificial intelligence and new technologies are causing rapid changes in urban planning, creating new functional urban spaces and changing the purpose of existing ones. Changes in the structure of locations and the recombination of functions are occurring under the spatial influence of innovations in ICT (Jacques et al. 2024). Such digital innovations in the field of transport, for example, are leading to the emergence of driverless transport/autonomous vehicles (AVs), which is leading to the repurposing of public parking spaces due to the reduction in the use of private cars (Batty 2024). However, as spatial interventions follow the application of new technologies, the algorithms that tell us how to optimally design cities need to be studied in more detail to ensure spatial equity for different groups of residents (Soydan and Oran 2025). In the field of transportation, this may mean citizens' right to equitable access to any important facilities, such as hospitals, business centers, or kindergartens (Hook et al. 2025).

Planning a public transport system involves several optimization criteria, which makes it difficult to select a globally optimal structure (Borndörfer et al. 2017). An effective system must meet the functional requirements of both passengers and operators, while ensuring accessibility and comfort for all residents (Borndörfer et al. 2017). Automating the planning process without using additional parameters (such as transport frequency schedules or taking into account the synergy of several modes of transport in a single network) can be considered as a TNDP (transport network design problem). This task is computationally complex due to the extensive search space and list of domain constraints (Mumford 2013). Given the complexity of the problem, researchers have applied various heuristic methods and agent-based modeling to find the optimal network configuration (Holliday and Dudek 2024).

Being in the field of mathematical problems, many works exploring TNDP solutions have been applied to synthetic graphs, with a few practical examples for different cities.

However, the transition from synthetic to real networks reveals serious problems that go beyond accelerating computations. From an urban planning perspective, a number of methodological questions arise regarding mathematically optimal solutions (Da et al. 2025). While AI reduces the time needed to find solutions for real cities (Holliday and Dudek 2024), synthetic tests (Mandl, Mumford) do not take into account the spatial heterogeneity of POI distribution and demographic complexity characteristic of real urban conditions (Rodríguez, Delgado-Lindeman, and dell'Olio 2025) to ensure a socially acceptable result.

Such solutions typically do not provide sufficient service to nodes with lower demand, while the balance between operator and customer costs can lead to noticeable shortcomings in customer service. Existing studies have shown that an algorithm priority aimed at complete customer satisfaction using the average travel time indicator lead to a huge increase in operating costs, making such solutions unfeasible in real-world conditions.

This paper answers the question of whether attempting to provide equitable access to facilities for a a bigger number of residents can be sustainable in terms of operator and customer costs, and whether there are any advantages to such an approach.

The contribution of this work is the introduction of an accessibility metric as a target optimization function based on the principle of fair accessibility. This approach has three

significant advantages. First, it provides a near-optimal solution, avoiding the search for the optimal balance of customer and operator weights. Second, the generated network demonstrates better resilience to disruptions. Third, although continuous accessibility itself remains close to the original solution, a more reliable transport network promises better spatial equality.

Related Work

The planning process in the field of public transport is conceptually divided into several stages, the results of which serve as input for the subsequent stage. Schmidt and Schöbel (2024) decompose this process into three major subproblems:

- Line generation,
- Line selection,
- Frequency setting.

In this work, we focus on the stages of line generation and selection, formally known as the line planning or Transit Network Design Problem (TNDP) (Mandl 1980; Durán-Micco and Vansteenwegen 2022; Şahin, Lindner, and Schlechte 2023). The objective of this stage is to determine a feasible set of transit lines that ensures convenient travel for passengers while minimizing operational costs. The infrastructure network is represented as a graph, where nodes correspond to stops and edges denote the physical or temporal connections between them, weighted by distance or travel time. Each transit line is represented as an ordered sequence of stops. The inputs to this problem typically include the underlying infrastructure network and a travel demand matrix, which is generally assumed to be static over the analysis period (Schmidt and Schöbel 2024). Extensions of this problem often include temporal or multimodal dimensions, such as frequency optimization (Jha, Jha, and Tiwari 2019) or rerouting strategies aimed at improving gravity-based accessibility without increasing operational costs (Rumpf and Kaul 2021).

A substantial body of research on TNDP relies on metaheuristic methods, which are well suited to handling its combinatorial nature and flexible objective formulations. Among these, evolutionary algorithms have gained particular prominence, including genetic algorithms designed with specialized operators for TNDP (Bagloee and Ceder 2011; Pternea, Kepaptsoglou, and Karlaftis 2015), and their multi-objective variants, such as NSGA-II (Liu et al. 2020), which balance passenger convenience and operator costs.

Recent research has increasingly explored the use of machine learning and neural network architectures capable of capturing the topological dependencies within transport graphs. In particular, graph neural networks (GNNs) and reinforcement learning (RL) methods enable modeling the process of route construction as a sequence of actions in a Markov decision environment. Several studies have demonstrated the potential of deep neural architectures in combinatorial optimization tasks structurally similar to TNDP, such as the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP), where GNN models are trained to generate graph-based solutions that account for spatial interdependencies (Khalil et al. 2017; Wu et al. 2021). Building upon these advances, Holliday and Dudek (2024) proposed integrating a GNN model trained in the RL paradigm with an evolutionary algorithm for the construction and improvement of public transport networks, showing that such hybrid approaches can effectively combine structural reasoning with adaptive optimization.

Despite notable progress in algorithmic methods for solving the TNDP, most studies remain primarily focused on improving computational efficiency and enhancing network-level or passenger performance metrics. Comparatively few works have addressed the question of how generated transit networks influence spatial accessibility and equity across urban territories.

Wang, Chau, and Araldo (2024) highlight the importance of incorporating spatial accessibility and geographic equity metrics into network optimization frameworks. Their model, based on Message Passing Neural Networks (MPNN) and reinforcement learning, aims to minimize disparities in access to urban opportunities, demonstrating that aligning network topology with the spatial and functional structure of the city can support the design of more equitable transport systems. However, such approaches remain limited, and the comprehensive evaluation of generated networks in terms of equal access to spatial units within the city is still an underexplored research direction.

Methodology

Our methodological framework is designed to compare different route generation strategies under a unified formulation of the Transit Network Design Problem. All candidate solutions are represented as feasible sets of routes on the same underlying city graph, subject to identical route length and connectivity constraints.

Each solution is evaluated using a composite objective that combines three components: passenger cost (C_p), operator cost (C_o), and a demand-weighted transport connectivity metric (C_w), complemented by penalty terms for infeasibility. By varying the weights assigned to these components, we emulate different planning perspectives (user-oriented, operator-oriented, and connectivity-oriented) while preserving a consistent evaluation scheme.

Within this common setting, we benchmark three algorithms, developed by Holliday and Dudek (2024): a learned constructive baseline (LC100), a metaheuristic baseline (BCO), and a neuro-evolutionary variant (NeuroBCO) that integrates a learned policy into the search process. All methods are run on the same benchmark instances with identical parameters, enabling a controlled assessment of how the choice of objective and optimization strategy affects network efficiency, connectivity, and the equity of access.

Problem Formulation

The Transit Network Design Problem is formulated on an augmented city graph

$$C = (N; E_s; D);$$

where N denotes the set of n potential transit stops (nodes), $E_s = f(i;j;\ _{ij})g$ is the set of street edges with associated travel times $_{ij} > 0$, and $D\ 2\ R^n{}_0{}^n$ is a symmetric demand matrix ($D = D^>$) specifying the number of trips between each origin–destination pair (Mandl 1980).

A *route* r is a sequence of distinct nodes connected by edges in E_s. The objective is to determine a set of routes $R = fr_1;:::;r_Sg$ that collectively form a feasible transit network, satisfying

$$\text{MIN} \quad jrj \quad \text{MAX};$$

for all $r\ 2\ R$, such that all nodes are mutually reachable via transit, and a multi-objective cost function is minimized. This cost typically balances passenger travel efficiency, operational expenditure, and network feasibility constraints. The TNDP is NP-hard due to the combinatorial nature of route selection and the interdependence of demand coverage and transfer penalties.

In this paper, we carry out experiments with generating sets of routes for several benchmark graphs with different configurations. The input consists of graphs where the vertices are public transport stops and the edges are paths between stops with weights in times (minutes). LC-100, EA, and NEA algorithms are used for generation (description of the algorithms is provided below). For each experiment, the following optimization functions were selected: Average Travel Time, Route Travel Time, and Weighted Transport Connectivity.

Neuro-evolutionary algorithm

The Neural-Evolutionary Algorithm (NEA) (Holliday and Dudek 2024) is a hybrid optimization method for the TNDP that integrates a reinforcement learning-based graph neural network (GNN) policy with an evolutionary algorithm (EA). The learned policy acts as a heuristic mutation operator, allowing the algorithm to explore the solution space more efficiently than purely random search strategies.

At the core of NEA lies a GNN policy (ajs), referred to as the *Learned Constructor*. It constructs a complete set of transit routes by interacting with an environment represented as a Markov Decision Process (MDP), where the city graph $C = (N;E_s;D)$ defines the state space. At each step, the policy decides whether to extend or terminate a route based on node and edge embeddings produced by a graph attention network. The cumulative reward corresponds to the negative of the overall cost function:

$$C(C;R) = \quad w_pC_p + (1 \quad)w_oC_o + \quad C_c; \qquad (1)$$

where C_p is the passenger cost, C_o the operator cost, and C_c a constraint penalty term. The parameters $\ , w_p, w_o$, and balance the influence of each component.

The passenger cost represents the average in-vehicle travel time across all demand pairs:

$$C_p(C;R) = \frac{\sum_{i;j} D_{ij}\ {}^R_{ij}}{\sum_{i;j} D_{ij}}; \qquad (2)$$

where ${}^R_{ij}$ denotes the shortest travel time between i and j along the transit network R, including a fixed penalty p_T per transfer.

The operator cost quantifies total route traversal time:

$$C_o(C;R) = \sum_{r\,2\,R} {}_r; \quad {}_r = \sum_{(u;v)\,2\,r} {}_{uv} + \sum_{(v;u)\,2\,r} {}_{vu}: \qquad (3)$$

The penalty term C_c accounts for disconnected node pairs and violations of route length constraints.

During evolutionary optimization, a population of candidate networks undergoes iterative mutation and selection. In addition to standard random mutations, NEA employs a *neural mutation*, where the GNN policy generates a new route to replace an existing one in the current solution. This hybridization combines the exploratory power of evolutionary search with the problem-specific knowledge embedded in the learned policy. The resulting framework achieves improved convergence and solution quality compared to classical EAs and standalone learned models, performing effectively on standard TNDP benchmarks.

Transport connectivity

The transport connectivity metric is based on the principle of mutual accessibility of urban areas and reflects how effectively the transport network enables movement between all pairs of spatial units (e.g., urban blocks). It is computed by determining the minimum travel time between all nodes in the intermodal transport network, normalized by the corresponding straight-line (Euclidean) distances (Morozov et al. 2023).

To account for the spatial distribution of demand, a demand-weighted component is introduced. Let $T^R\ 2\ R^{n\ n}$ denote the matrix of all-pairs shortest transit times. The weighted median connectivity is then defined as:

$$C_w(C;R) = \frac{1}{n} \sum_{i=1}^{n} \text{median} \quad T^R_{ij} \quad \frac{D_{ij}}{\max_k D_{ik}} \quad T^R_{ij} < 1 \quad : \qquad (4)$$

This formulation emphasizes accessibility to high-demand destinations and complements traditional cost-based optimization criteria by linking the structure of the transport network with the spatial distribution of travel needs, allowing for a more comprehensive evaluation of urban connectivity.

Experiments and Results

Data

All **experiments**[1] were conducted on the standard public transit benchmarks (Mumford, C. L. 2014): the Mandl network (15 nodes, 6 routes) and the four larger Mumford instances (ranging from 30 to 127 nodes and 12 to 60 routes), as detailed in Table 1. These synthetic cities vary significantly in scale, demand distribution, and spatial structure, providing a robust testbed for evaluating transit network generation algorithms under diverse conditions. The algorithms under comparison - LC-100, BCO, and the hybrid NeuroBCO - were executed with a fixed random seed and identical route length constraints to ensure reproducibility and fair performance assessment.

[1] https://anonymous.4open.science/r/TNDP_learning-EC76/experiments.ipynb

Table 1: Statistics of the five synthetic benchmark cities used in our experiments.

City	n	jE_sj	S	min/max	Area (km^2)
Mandl	15	20	6	2 / 8	352.7
Mumford0	30	90	12	2 / 15	354.2
Mumford1	70	210	15	10 / 30	858.5
Mumford2	110	385	56	10 / 22	1394.3
Mumford3	127	425	60	12 / 25	1703.2

Assessment

Following the framework proposed by Šfiligoj, Peperko, and Cats (2025), we adopt a set of indicators derived from the temporal evolution of the *access graph*, a representation in which two public transport stops are connected if the generalized travel time between them is below a certain threshold. These indicators capture both the dynamics and the equity of accessibility in a network as the travel time budget increases. Specifically, the parameters t_M, t_M, D_M, D_{30}, G_M, and G_{30} quantify the characteristic time at which the network becomes substantially connected, the relative efficiency of this process, the corresponding reachability levels, and the distributional equity of access. This set of metrics allows for a consistent, comparative evaluation of public transport network performance in terms of accessibility and robustness:

- t_{max}: the maximum travel time considered on the network.
- t_M: characteristic median travel time at which network connectivity rapidly increases.
- t_M: variability in travel time around t_M, capturing network temporal heterogeneity.
- D_M and D_{30}: average degrees of the access graph at median travel time and at 30 minutes, reflecting reachability.
- G_M and G_{30}: Gini coefficients at the respective times, measuring inequality of access across nodes.
- Algebraic Connectivity: spectral measure of network robustness and connectivity linked to overall accessibility equity.

In all experiments, we fix the average waiting time to 5 minutes, the waiting-time weight to 1 minute, and the time-budget discretization step to 2 minutes.

Experiments on Benchmark Data

The core of our analysis revolves around the demand-weighted connectivity metric C_w, which integrates spatial accessibility, travel efficiency, and demand responsiveness into a single objective. Unlike traditional metrics that focus solely on average travel time (C_p) or total route length (C_o), C_w captures how effectively the network connects urban areas in proportion to real travel needs. Results across all benchmarks (Tables 1–5 in additional materials, consolidated in Table 6 in additional materials) reveal that C_w acts as a natural mediator between passenger and operator objectives, preventing extreme trade-offs and promoting more balanced, equitable transit systems.

When optimization prioritizes passenger convenience (high C_p weight), networks achieve the lowest average travel times by constructing dense, overlapping routes around high-demand corridors. While this minimizes in-vehicle time and transfer penalties, it comes at a steep operational cost: C_o often increases by 2-3 times compared to cost-focused solutions. For example, on the Mandl benchmark, passenger-optimized configurations reduce C_p to approximately 10.4 minutes but inflate C_o beyond 350 minutes of total route traversal. This pattern scales with network size-on Mumford3, C_o exceeds 8700 under passenger-focused settings-highlighting the economic unsustainability of purely user-centric designs.

In contrast, minimizing operator cost generates compact, tree-like structures with minimal route mileage. These networks are highly efficient from an operational standpoint, achieving the lowest C_o across all instances, but at the expense of passenger experience. Direct-trip shares (d_0) plummet - often below 15% in larger networks - and multi-transfer journeys dominate, with over 30% of trips requiring two or more changes in Mumford3. Such configurations create accessibility bottlenecks, particularly in low-demand peripheral areas, resulting in spatially inequitable service.

The introduction of C_w as a primary or balanced objective fundamentally alters this dynamic. Networks optimized with significant C_w weighting exhibit substantially improved transfer profiles: direct trips rise sharply (up to 70% in balanced settings on smaller graphs), and long transfer chains are nearly eliminated (see Figure 3 in additional materials). On Mumford0, for instance, d_0 increases from just 15.5% under operator minimization to nearly 70% in connectivity-aware configurations, with two-transfer trips dropping to under 1%. This enhancement in direct connectivity occurs without fully sacrificing efficiency - C_o remains within 20-50% of the operator-optimal baseline in most cases.

Balanced weight combinations (e.g., equal or moderate emphasis on C_p, C_o, and C_w) consistently produce solutions near the Pareto front, achieving near-minimal C_w while preserving competitive values in both C_p and C_o. These configurations demonstrate the complementary nature of the three objectives: connectivity-driven routing naturally induces redundancy that benefits passengers (shorter paths, fewer transfers) while avoiding the excessive sprawl of pure passenger optimization. Figure 1 and 2 in additional materials illustrates this synergy, showing normalized metric trade-offs across weight settings for NeuroBCO on Mumford3.

Among the tested algorithms, NeuroBCO demonstrates superior performance when C_w is included in the objective. It consistently achieves the lowest C_w values - often by margins of 5–8% on larger instances - while maintaining strong C_p scores. This advantage stems from its hybrid architecture, which combines learned graph-aware construction with evolutionary refinement, enabling more nuanced exploration of demand-structured connectivity patterns than purely heuristic or evolutionary baselines.

Despite these benefits, over-emphasizing C_w reveals important limitations. Exclusive connectivity optimization increases C_o by 20–40% relative to balanced settings, as the